`tessera`: An R package for detecting regime boundaries in ecological systems using similarity network topology

João Tonini[1*]

[1]Department of Biology, University of Richmond, Richmond, VA 23173, USA

[*]Corresponding author: jtonini@richmond.edu

ORCID: 0000-0002-4730-3805

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Abstract

1. Detecting regime shifts in ecological systems is critical for conservation and management, yet traditional approaches rely on univariate thresholds or feature-space classifiers that discard information encoded in the relational structure among entities. When ecological communities undergo regime shifts, the topology of how entities relate to each other changes in ways that standard methods cannot capture.

2. Here I present `tessera`, an R package implementing the TESSERA framework (Topology-Enhanced Similarity and Structure for Entity Regime Analysis). `tessera` detects regime boundaries by computing multi-measure entity similarity, constructing networks, identifying community structure, and quantifying how that structure diverges across temporal bins. The package provides a pipeline of inspectable objects, domain-specific presets, sparse representations for scalability, and builds on established ecology R infrastructure (`igraph`, `vegan`, `ape`).

3. I demonstrate `tessera` using the Global Coral Bleaching Database (25,154 records after quality filtering, 1983–2019), showing that similarity network topology reveals a pattern of monotonically increasing regime divergence across the study period. Community structure progressively fragments—from 20 communities with modularity 0.81 in 1983–2003 to 15 communities with modularity 0.75 in 2013–2019—consistent with chronic environmental drift rather than a single sharp regime shift.

4. `tessera` is available on CRAN and GitHub (`https://github.com/jtonini/tessera-r`) under an MIT license. It requires no Python dependencies and is installable with `install.packages("tessera")`.

Keywords: regime shift, similarity network, community detection, coral bleaching, temporal dynamics, ecotone

1 Introduction

Regime shifts—abrupt transitions between alternative stable states—are among the most consequential phenomena in ecology (Scheffer et al., 2001; Folke et al., 2004). Detecting the boundaries where these transitions occur, termed ecotones in biogeographical tradition (Clements, 1905), remains methodologically challenging, particularly in high-dimensional observational datasets where regime boundaries are not aligned with simple thresholds in individual variables.

Existing approaches to regime detection in ecology fall into two broad categories. Threshold-based methods, such as sequential regime shift detection (Rodionov, 2004) and changepoint analysis (Killick & Eckley, 2014), operate on univariate or low-dimensional summaries and cannot capture multivariate structural shifts. Multivariate classifiers such as random forests and gradient-boosted models operate in feature space but treat entities independently, discarding the relational structure—which entities are similar to which—that encodes the topology of the system.

Biogeographical theory suggests an alternative perspective. Ecotones are defined not by the properties of individual organisms but by the topology of community structure: the pattern of which entities cluster together and how those patterns shift across space or time (Uyeda & Harmon, 2014). When a system crosses a regime boundary, the similarity relationships among entities reorganize. This topological signal is invisible to feature-space classifiers but is detectable through network analysis.

The TESSERA framework (Topology-Enhanced Similarity and Structure for Entity Regime Analysis; Tonini, in preparation) formalizes this insight into a computational pipeline. It computes pairwise entity similarity using multiple ecologically motivated measures, constructs similarity networks, detects community structure, and quantifies how that structure diverges across temporal bins. The magnitude and pattern of this divergence—regime divergence (RD)—reveals both the location and the geometry of regime boundaries.

Here I present `tessera`, an R package that implements the analytical core of the TESSERA

framework (Steps 1–3: Characterize, Discover, Quantify) natively in R. The package is designed for ecology researchers: it builds on established R infrastructure (`igraph`, `vegan`, `ape`), uses sparse representations for scalability, provides domain-specific presets, and returns inspectable objects at every pipeline step. I demonstrate the package using the Global Coral Bleaching Database (van Woesik et al., 2022) and discuss its application to ecological regime detection.

2 Description of the `tessera` Package

2.1 Design philosophy

`tessera` is structured as a pipeline of five core functions, each producing an S3 object with `print()`, `summary()`, and `plot()` methods (Figure 1). This design allows researchers to inspect intermediate results at every stage rather than treating the analysis as a black box.

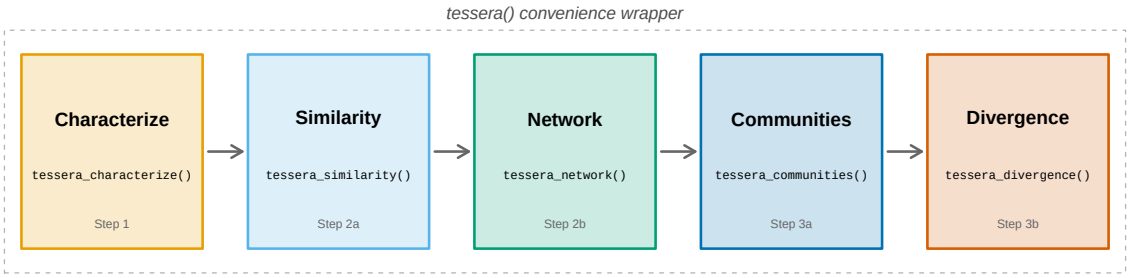

Figure 1: The `tessera` pipeline. Data flows through five functions, each producing an inspectable S3 object. Temporal binning (Step 1) feeds into multi-measure similarity computation (Step 2a), which is sparsified and converted to network form (Step 2b). Community detection (Step 3a) partitions the network, and regime divergence (Step 3b) quantifies how community structure changes across temporal bins. The convenience wrapper `tessera()` executes all steps with domain-specific presets.

Three design decisions reflect lessons from applying the framework to large ecological datasets. First, similarity matrices use sparse k-nearest-neighbour (kNN) representation by default via `Matrix::sparseMatrix`. For the coral bleaching dataset (26,962 records), a

dense similarity matrix requires approximately 5.8 GB of memory; the sparse kNN representation with $k = 15$ requires approximately 5 MB—a reduction that makes the analysis tractable on standard hardware. Second, temporal binning uses quantile-based boundaries by default rather than fixed-width intervals, ensuring balanced entity counts across bins. This is critical for long-span ecological datasets where early observation periods are sparsely sampled. Third, the package is fully native R with no Python dependency, ensuring that `install.packages("tessera")` is the only step required.

2.2 The pipeline

The five core functions correspond to TESSERA's analytical steps:

`tessera_characterize()` (Step 1: Characterize) assigns entities to temporal bins, scales features within bins to zero mean and unit variance, handles missing values, and merges bins with too few entities. Key parameters include `n_bins` (default 4), `bin_method` (`"quantile"` or `"fixed"`), and `min_entities` (default 30).

`tessera_similarity()` (Step 2a: Discover) computes pairwise entity similarity within each temporal bin using one or more of seven measures. Three are ecologically standard: Bray-Curtis dissimilarity (via `vegan`; Oksanen et al., 2022), Jaccard similarity, and Pearson correlation. One is specific to TESSERA: Simpson similarity, which discretizes each feature into quantile bins and computes the proportion of features for which two entities share a bin. Three are general-purpose: cosine, Euclidean, and Mahalanobis. The output is sparsified to retain only the k most similar neighbours per entity, with $k = 15$ as the default.

`tessera_network()` (Step 2b: Discover) constructs similarity networks from the sparse similarity matrices via kNN (default), threshold, or weighted methods, returning `igraph` objects (Csárdi & Nepusz, 2006) with similarity as edge weights.

`tessera_communities()` (Step 3a: Quantify) detects community structure using one of four algorithms: Leiden (via the `leiden` package; Traag et al., 2019), Louvain (built into `igraph`), spectral clustering (via `kernlab`; Karatzoglou et al., 2004), or Neighbour-Joining

(via `ape`; Paradis & Schliep, 2019). The NJ option provides a direct bridge to phylogenetic methods familiar to ecologists. Community count is capped at 20 by default to prevent feature-space drowning in downstream analyses.

`tessera_divergence()` (Step 3b: Quantify) measures how community structure changes between temporal bins using Normalized Mutual Information (NMI), Variation of Information, or a composition-shift metric. The function supports adjacent-bin, first-bin, and all-pairs reference strategies, and returns per-entity divergence contributions that identify which entities drive regime shifts.

2.3 Convenience wrapper and presets

The function `tessera()` runs the full pipeline in a single call. Named presets encode domain-appropriate parameter combinations: the `"ecology"` preset uses Bray-Curtis and Simpson similarity with Leiden community detection, reflecting standard ecological practice. Parameters specified by the user override preset defaults, preserving full configurability.

```
library(tessera)
result <- tessera(coral_data, time_col = "year",
                  preset = "ecology")
```

2.4 Dependencies and installation

`tessera` depends on `igraph` ($\geq$1.5.0), `vegan` ($\geq$2.6), `ape` ($\geq$5.7), `Matrix`, and base R packages. Optional dependencies in `Suggests` include `leiden`, `kernlab`, and `ggplot2`. The package is available from CRAN and from GitHub at `https://github.com/jtonini/tessera-r`.

3 Application: Coral Bleaching Regime Detection

3.1 Data

I demonstrate `tessera` using the Global Coral Bleaching Database (van Woesik et al., 2022), a publicly available dataset (figshare, CC-BY 4.0). The database was queried by joining sample event records with environmental covariates, yielding 26,962 records spanning 1983–2019. After removing records with missing values in any of the nine environmental features, 25,154 complete records remained. Features include sea surface temperature (SST mean), SST anomaly (SSTA), SSTA degree heating weeks (DHW), SSTA frequency, thermal stress anomaly (TSA), TSA DHW, climatological SST, depth, and windspeed.

3.2 Analysis

I applied the full `tessera` pipeline with the ecology preset. Quantile-based temporal binning ($n = 4$) produced four bins with approximately balanced entity counts: bin 1 (1983–2003, $n = 6{,}331$), bin 2 (2004–2006, $n = 6{,}355$), bin 3 (2007–2012, $n = 6{,}485$), and bin 4 (2013–2019, $n = 5{,}983$). Similarity was computed using Bray-Curtis and Simpson measures with sparse kNN ($k = 15$). Community detection used the Louvain algorithm (resolution = 1.0), and regime divergence was quantified using NMI with adjacent-bin reference. The full pipeline completed in 61 seconds on a single core.

```
result <- tessera(coral_data,
                  time_col = "year",
                  preset = "ecology")
```

3.3 Results

Regime divergence was high across all three bin transitions and increased monotonically: 0.924 (bin 1→2), 0.938 (bin 2→3), and 0.944 (bin 3→4; Figure 2). This pattern corresponds

to the diffuse geometry in the TESSERA framework, indicating continuous reorganization of community structure rather than a single sharp regime boundary.

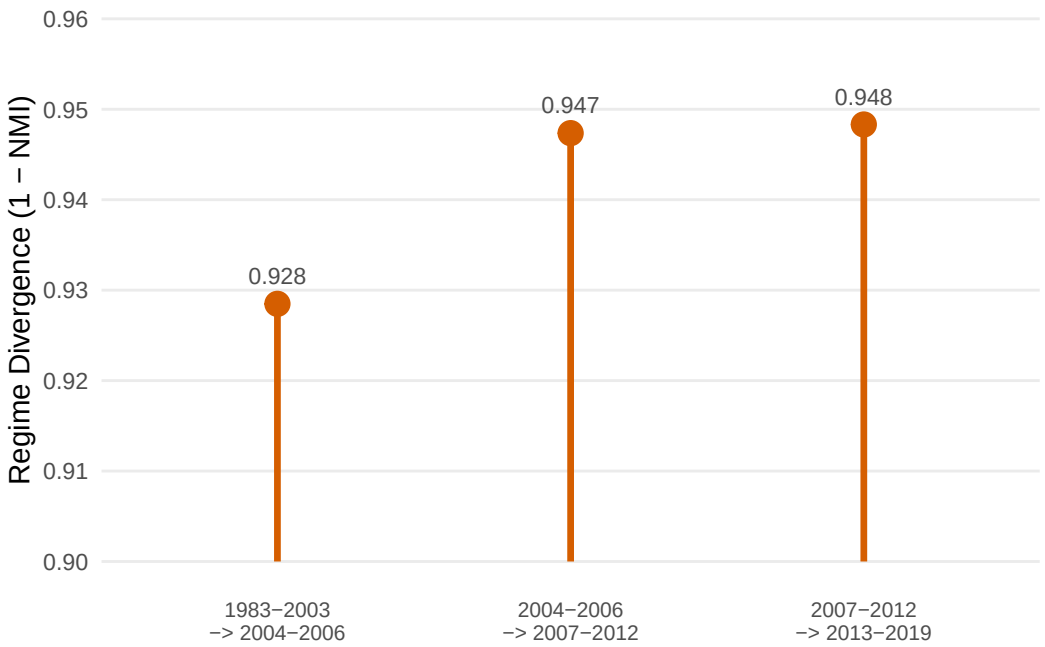

Figure 2: Regime divergence (1 − NMI) across adjacent temporal bins for the Global Coral Bleaching Database (25,154 records, 9 environmental features). Divergence increases monotonically from 0.924 (1983–2003 → 2004–2006) to 0.944 (2007–2012 → 2013–2019), consistent with a diffuse regime divergence geometry indicating continuous environmental restructuring. Note the truncated y-axis; all values fall within the narrow range 0.92–0.95.

Community detection revealed a progressive fragmentation of network structure across the study period. The earliest bin (1983–2003) contained 20 communities with a modularity of 0.806, indicating well-defined clusters of environmentally similar reef observations. Community count decreased to 15 by the final bin (2013–2019) while modularity declined to 0.752 (Figure 3). The intermediate bins showed a parallel pattern: bin 2 (2004–2006) had 20 communities with the highest modularity (0.836), while bin 3 (2007–2012) dropped to 16 communities and a modularity of 0.745. This decline in both community count and modularity indicates that the similarity network became less structured over time—reef observations that previously clustered into distinct environmental regimes became more homogeneously distributed across the network.

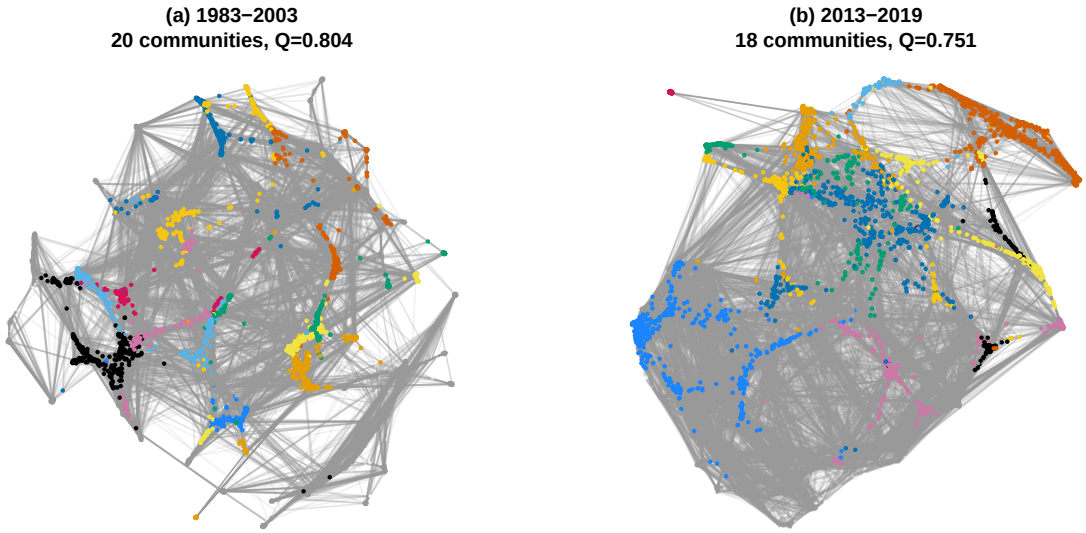

Figure 3: Similarity network community structure for two temporal bins spanning the full study period: (a) 1983–2003 (20 communities, modularity $Q = 0.806$) and (b) 2013–2019 (15 communities, $Q = 0.752$). Nodes represent reef observations; edges connect entities within each other's k-nearest neighbours ($k = 15$, Bray-Curtis similarity); colours indicate community membership (Louvain algorithm, Okabe-Ito palette). The reduction in community count and modularity between bins reflects progressive fragmentation of environmental regime structure.

Per-entity divergence analysis revealed that 97.9% of entities in the largest bin changed community membership across at least one temporal transition (Figure 4). The mean divergence among entities with non-zero scores was 0.920, and 100% of entities achieved the maximum divergence score of 1.0 in at least one transition. This near-universal community reassignment underscores the pervasiveness of the topological reorganization.

3.4 Interpretation

The monotonically increasing regime divergence and progressive loss of community structure tell a coherent ecological story. Rather than revealing a single regime shift at the 1998 mass bleaching event (Hughes et al., 2018), the analysis detects a continuous process of topological reorganization spanning the entire study period. This diffuse geometry is consistent with the interpretation that coral reef environmental regimes have been undergoing

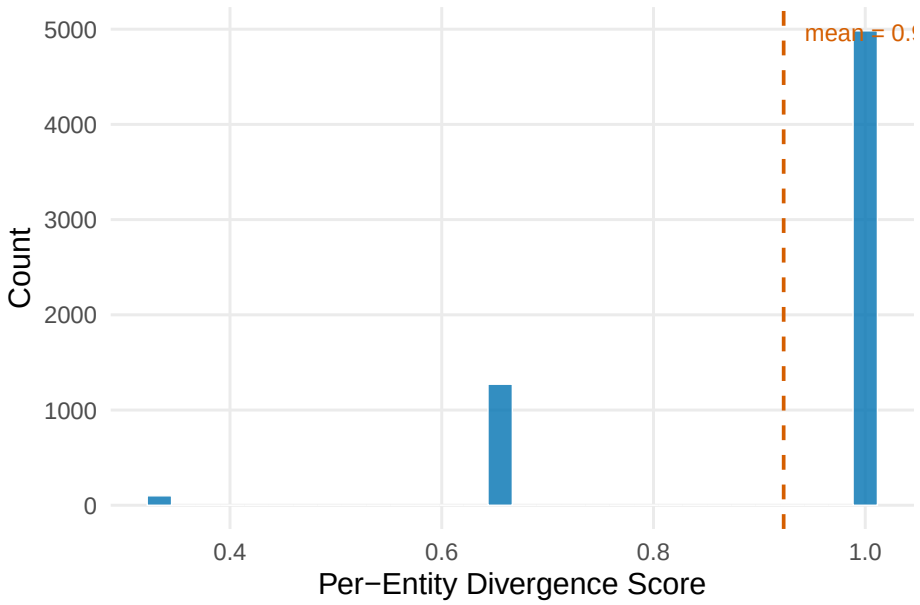

Figure 4: Distribution of per-entity divergence contributions across reef observations in the largest temporal bin ($n = 6{,}485$). Of these entities, 97.9% changed community membership across at least one bin transition, with a mean divergence score of 0.920 (dashed line). The concentration of scores near 1.0 indicates near-universal community reassignment, reflecting system-wide topological restructuring rather than localized regime shifts at the margins.

sustained restructuring driven by cumulative thermal stress, increasing SSTA frequency, and the compounding effects of repeated bleaching events.

The declining modularity across bins is particularly informative. High modularity in the early period (0.806–0.836) indicates that reef observations clustered into distinct, well-separated environmental regimes—groups of reefs with internally coherent thermal, depth, and windspeed profiles. The erosion of this structure by the final bin (modularity 0.752) suggests that environmental conditions are becoming more homogeneous in their stress profiles, or alternatively, that the diversity of distinct environmental regimes is collapsing.

Critically, this topological signal is not accessible through standard feature-space classification. A model predicting bleaching from SST and depth treats each reef independently. The similarity network, by contrast, encodes which reefs behave similarly under comparable environmental conditions—and it is the pattern of this similarity, not the feature values themselves, that reorganizes. Two reefs with identical SST anomalies may fall into differ-

ent similarity communities because their broader environmental contexts (depth, windspeed, prior thermal history) place them in different network neighbourhoods. The near-universal entity reassignment (97.9%) demonstrates that this is not a localized phenomenon affecting a few marginal reefs, but a system-wide topological restructuring.

4 Discussion

`tessera` provides ecologists with a new analytical approach to regime detection that operates on a fundamentally different signal than existing methods. Where changepoint analysis (Killick & Eckley, 2014) detects shifts in distributional parameters and multivariate classifiers detect decision boundaries in feature space, `tessera` detects changes in the topology of entity similarity networks. These approaches are complementary: a system may exhibit a topological regime shift (reorganization of which entities cluster together) without a corresponding shift in mean feature values, and vice versa.

The package is not a workflow assembling existing methods into a pipeline. While it builds on established R infrastructure—`igraph` for network operations, `vegan` for ecological dissimilarities, `ape` for phylogenetic methods—the analytical contribution is the specific use of similarity network topology as a regime detection signal. The Simpson similarity measure (quantile-bin overlap), the regime divergence quantification, the three failure geometries, and the per-entity divergence attribution are novel to this framework.

The current R implementation covers TESSERA's analytical core (Steps 1–3: Characterize, Discover, Quantify). The prediction layer (Steps 4–6), which uses graph neural networks, long short-term memory networks, and ensemble meta-learning to forecast regime transitions, is available in the Python implementation (`tessera-ml` on PyPI) and is planned for a future R release using R `torch`. This division is deliberate: the analytical framework in `tessera` stands alone as a contribution to regime detection methodology, while the prediction layer extends it to forecasting.

Several limitations should be noted. Similarity measure ranking is domain-dependent—Bray-Curtis and Simpson perform well for ecological data, but other domains may require different measure combinations. The ecology preset provides sensible defaults, but researchers should evaluate measure performance for their specific systems. Temporal binning assumes that the relevant dynamics operate on the timescale captured by the bins; for systems with irregular or event-driven transitions, adaptive binning strategies may be needed. Adaptive fusion of multiple similarity measures (learned convex combination) is a planned extension.

4.1 Data and code availability

`tessera` is available on CRAN (`install.packages("tessera")`) and GitHub (`https://github.com/jtonini/tessera-r`) under an MIT licence. The coral bleaching dataset is bundled as `coral_example` (500-record stratified subset) and the full database is available from figshare (doi:10.6084/m9.figshare.20078993; van Woesik et al., 2022). All analyses reported here are reproducible via the package vignette.

Acknowledgements

[To be added.]

Author Contributions

J.T. conceived the TESSERA framework, developed the R package, performed the analyses, and wrote the manuscript.

Conflict of Interest

The author declares no conflict of interest.

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